

Name: _____

Class: _____

ACTIVITY SHEET

Chapter 3: Prior knowledge

- 1 What makes atoms electrically neutral (i.e. uncharged)?

- 2 Are all electrically neutral atoms chemically stable (i.e. unreactive)?

- 3 What makes atoms chemically stable?

- 4 Which electrically neutral atoms are chemically stable?

- 5 How can other atoms become chemically stable?

- 6 What is a chemical element?

- 7 Do metal or non-metal atoms tends to take electrons off other atoms? Explain.

8 Do metal or non-metal atoms tends to lose electrons to other atoms? Explain.

9 What are ions?

10 What type of ions do non-metal atoms form and why?

11 What type of ions do metal atoms form and why?

12 What happens to the metal and non-metal ions that are formed?

13 How does ionic bonding result in a regular array (lattice) of atoms?

14 What is a compound and what type of compound is held together by ionic bonds?

15 How do the atoms of a metal share electrons?

16 How does this result in metallic bonding?

17 What physical properties of metals result from this metallic bonding?

18 How do non-metal atoms share electrons?

19 Do only identical non-metal atoms share electrons or can non-identical ones do so as well?

20 How does this result in covalent bonding?

21 What two types of compounds are held together by covalent bonding?

Having difficulty?

- Visit the BBC Bitesize website on 'Ionic bonding' and explore page 1.
- Visit the BBC Bitesize website on 'Covalent bonding' and explore page 1.
- Visit the BBC Bitesize website on 'Metal structures and their properties' and explore pages 1 and 2.